
📄 Part A — Theory: Classification in Machine Learning (Final Short Version)

1. Introduction to Classification

What is Classification?

Classification is a supervised machine learning technique used to predict **categorical labels** from input data. The model learns from labeled data and assigns new data into classes such as **Yes/No or Approved/Rejected**.

Mathematically:
$f(X) \rightarrow Y$, where Y is a class label.

Example: predicting whether a loan is **Approved or Rejected** based on applicant data.

Classification vs Regression

- **Classification:** Predicts categories (Yes/No)
- **Regression:** Predicts numbers (e.g., price)

Examples:

- Spam detection → Classification
- House price prediction → Regression

2. Classification Algorithms

Logistic Regression

Uses a sigmoid function to predict probability between 0 and 1.

- Output ≥ 0.5 → Class 1
- Output < 0.5 → Class 0

Use: Loan approval, medical diagnosis
Pros: Simple, fast, interpretable
Cons: Poor for complex patterns

Decision Tree

Splits data based on feature conditions until a decision is reached.

Use: Credit scoring, customer classification
Pros: Easy to understand, no scaling needed
Cons: Overfitting risk

Random Forest

An ensemble of multiple decision trees using majority voting.

Use: Fraud detection, loan prediction
Pros: High accuracy, reduces overfitting
Cons: Less interpretable, slower

3. Classification Metrics

Accuracy

Correct predictions / total predictions

Precision

TP / (TP + FP) → correctness of positive predictions

Recall

TP / (TP + FN) → ability to find actual positives

F1-Score

Balance of Precision and Recall

Confusion Matrix

Actual \ Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Metric Summary

- Accuracy → overall correctness
- Precision → avoid false positives
- Recall → avoid false negatives
- F1 → balanced performance

4. Class Imbalance

Accuracy can be misleading when one class dominates.

Example:

- 90% loans rejected
- Model predicts all “Rejected” → 90% accuracy but useless

Loan Example:

- Precision → important for safe loan approval
- Recall → important for not rejecting good customers
- F1-score → best balance

5. Real-World Case Study: Fraud Detection

Goal

Detect fraudulent transactions in banking systems.

Data

Transaction amount, time, location, user history.

Models

Logistic Regression, Random Forest, Neural Networks.

Key Insight

- Dataset is highly imbalanced
- Recall is very important
- Ensemble models perform better

Conclusion

Classification is widely used in real systems like finance and healthcare. Models such as Logistic Regression, Decision Tree, and Random Forest have different strengths. Proper evaluation using Precision, Recall, and F1-score is essential, especially in imbalanced datasets like loan approval.
